HCXAI & RAI Semester Project

Problem → People → Needs → Design → Evaluation

Project Description

The Semester Project is where you put HCXAI into practice. In teams of 3–4, start with a consequential problem in XAI, identify the people who encounter it, understand what they need, design something in response, and evaluate what happens when people actually engage with that design. **The problem may include a Responsible AI dimension, but the project itself must approach that problem through a HCXAI lens.**

A strong project should make the full chain of reasoning visible: why this problem, why these people, what you learned from them and from the literature, why that led to this design, how you evaluated it, and what you learned from the result. **Every project must meaningfully engage real people or reasonable proxies, produce a concrete design response, and evaluate that response using mixed methods with a meaningful qualitative component.** The design can be lightweight and the evaluation can be formative. What matters is that the problem is scoped intelligently, the methods fit the question, the design follows from evidence rather than assumption, and the team is honest about limitations, failures, and what should change next. **The project does not need to be conference- or peer-review-ready. It does need to be rigorous.** If you want to take it further as a portfolio piece, internship/job-interview artifact, or future research project, you are welcome to plan accordingly.

Recall that **CS 7375 is a PhD-level HCI seminar**, so there is an assumption of basic familiarity with HCI concepts and research methods. You do not need to be an HCI specialist. If HCI is newer to you, use this as an opportunity to build those muscles. Learn from each other, independently fill in gaps, read beyond the assigned material when needed, and spend extra time getting comfortable with unfamiliar methods/concepts. It will pay off in the long run. *Use the project to stretch beyond the methods and skills you already know.*

The Human-centered Design Requirement

Do not invent user needs from your desk.

Every project must meaningfully engage real people. This can include your lab mates, classmates, friends, colleagues, domain experts, potential users, or reasonable proxy participants. Literature can help identify likely needs; direct engagement should help test, refine, or challenge those assumptions. Synthetic users and LLM-generated participant responses do not count as human engagement and cannot be presented as human data.

- **Understand the need.** Who are the relevant people? What are they trying to accomplish? Where does explanation, transparency, oversight, contestability, or accountability matter? What do they actually need?
- **Design in response.** What did you design or redesign because of what you learned? How did prior work and human input shape the choices?
- **Evaluate and learn.** What are you evaluating, what are you evaluating it against, and what did people actually experience?

A technical implementation, literature review, survey, or interview study alone is not enough. The understanding phase must lead to a design response, and that design must be put back in front of people.

What Counts as a Design Response?

Your design response could be:

- **a low-, mid-, or high-fidelity interactive prototype**
- **an interactive scenario or speculative interface**
- **a toy system**
- **several competing XAI designs that participants can compare or react to**

The point is to create something concrete enough for people to interact with, react to, and evaluate. **This is not a software-engineering assignment.** Build only as much as you need to investigate the human-centered question. See the Sample Project walkthrough to get an overall idea.

Rigor Is in the Process

Rigor is in the process, not the polish of the output. I am looking for a well-grounded problem, thoughtful engagement with people, defensible design choices, meaningful evaluation, and honest reflection on what you learned.

Scope down before you scope up. Choose a problem around the people you can realistically reach, the skills available on your team, the literature you can engage, and an evaluation you can actually complete.

Failure and negative results are okay. If the design does not work, show what happened, ask participants why, explain what assumption broke, and tell us what you would change next.

If you want the project to become a portfolio piece, internship artifact, workshop submission, or job-interview example, plan accordingly. Projects from this kind of work have traveled beyond the classroom before. That is welcome, but it is above the minimum bar for the course.

Evaluation

Every project needs a meaningful evaluation. Think mixed methods: every evaluation must include a meaningful qualitative component, while quantitative evidence can complement it where useful.

Possible evidence includes think-alouds, interviews after use, open-ended responses, structured tasks, ratings, task success, completion time, comparisons between designs, confidence/calibration measures, or other evidence justified by the question.

There is no magic sample size. For a small formative class project, roughly 6–10 participants can be reasonable. Match the sample to what you are trying to learn and the claims you want to make.

Be explicit about what you are evaluating, what you are evaluating it against, why that comparison makes sense, and what counts as success. "People liked it" is not enough. Aim to complete a real evaluation. If part of it does not work out, show what you tested, explain what blocked or weakened it, and give a credible next-round plan.

Reciprocal Participation Credit

Reciprocal Participation Credit (optional): Substantive **team-to-team reciprocal participation** in one another's evaluations can contribute **bonus points to the Evaluation portion of the final project rubric.** Participation should involve meaningful engagement with the other team's study and should be briefly documented in the final report. Teams that do not participate are not penalized.

Starter Pack: A Few Ways In

These are provided for ideation purposes, not as exhaustive tracks. Bring your own problem if you already have one.

Human-centered mechanistic interpretability. Who is the interpretation for, what decision should it support, and what form would make it useful or actionable?

Start from your own research. Find a real explanation, oversight, transparency, contestability, or responsibility problem in a domain you know, and ask **how HCXAI could help address it**. Use your expertise as an entry point, but do not simply repackage existing work.

Creativity, music, and writing. Explore explanations around generation, recommendations, editing, provenance, creative control, or intention mismatch.

Productivity and work. Ask what people need to understand about what an AI tool did, missed, or why they should intervene.

Agentic AI and oversight. What should people be able to see, understand, question, stop, or contest when AI begins taking actions?

Stress-test existing XAI. Take an existing technique or system, find the gap between what it provides and what people need, and redesign it.

Teams

The project will be completed in teams of 3–4 students. Teams of 3 are strongly preferred. No teams of 2. Aim for complementary teams. A strong team might combine technical/development experience, Design/HCI, empirical research methods, domain knowledge, policy/governance, or other relevant expertise.

Every member should own a load-bearing intellectual contribution to the project. Detailed team formation happens through the Canvas discussion. By **October 10**, the team needs to be final; the project idea does not. If you remain unassigned after the deadline, you will be assigned to the remaining available team or students. That assignment will be final, and you may need to work on a problem that the team has already chosen.

Sample Project Walkthrough: Human-Centered Mechanistic Interpretability

Suppose a team is interested in mechanistic interpretability. Instead of asking only whether a method can identify a feature or circuit, the team asks **what someone could actually do with that information**. They focus on **model debugging** and talk to ML/NLP researchers, developers, or relevant lab mates about how they investigate unexpected model behavior, what evidence they trust, and what would change their next action. Based on those needs, the team creates **2-3 interactive designs** that present the same mechanistic evidence differently, for example a raw technical representation, a task-oriented explanation, and one that makes uncertainty or evidentiary strength more visible. Participants then use those designs in a concrete debugging scenario. The team evaluates what people understand, misunderstand, decide to investigate next, and how confident they are, **combining qualitative feedback with quantitative measures where useful**. The contribution **does not need to be a new mechanistic-interpretability algorithm**; it can show how mechanistic evidence should be presented or qualified to better support a human task. **If the design fails, that is still useful evidence** if the team can explain what happened, what it learned, and what should change next.

Project Timeline

- **October 10:** Team formation complete. The team needs to be final. The project idea does not.
- **October 12–16:** Scope the project. Meet regularly, explore the literature, narrow the problem, identify the people you can realistically reach, and begin shaping the design direction.
- **October 19:** Quick in-class update, maximum 3 minutes per team. Cover who you are, the problem or use case, the relevant people, what the preliminary literature is telling you, what you currently think the contribution/design could be, and what you will complete before Oct. 26.
- **October 26:** Mid-Semester Project Checkpoint. Maximum 7 minutes presenting + 5 minutes Q&A. No slide-count rule.
- **November 30:** Final project presentation and submission.

Mid-Semester Project Checkpoint

The checkpoint is a live, formative progress check. The goal is to make the developing project visible while there is still time to improve, redirect, or strengthen it. Polish is not the goal. A thoughtful pivot can count as strong progress if you can explain what you learned and why the project changed.

What to Cover

- **Problem + grounding:** What problem are you pursuing, why does it matter, and what existing work are you building from or challenging?
- **People + needs:** Who have you engaged with so far, and what are you learning about goals, expertise, context, breakdowns, or explanation needs?
- **Design response + rationale:** What are you designing or redesigning because of what you learned, and why does that response make sense?
- **Evaluation + method fit:** What exactly will you evaluate, what will you evaluate it against, who will participate, and how will meaningful qualitative evidence be part of the plan?
- **Rigor + path forward:** What is still uncertain, what happens next, and what does each person own?

Checkpoint Rubric (10 points)

Criterion	2 - Strong	1 - Developing	0 - Not yet
Problem + Grounding	Clear, consequential HCXAI/RAI problem with appropriate scope and meaningful grounding in relevant prior work.	Promising direction, but the problem, scope, or grounding still needs work.	Problem is unclear, over-scoped, or mostly generic AI.
People + Needs	Relevant people are clearly identified, and literature and/or preliminary human engagement is producing concrete needs that matter to the project.	Relevant people are identified, but needs are still broad, weakly supported, or partly assumed.	Human needs are mostly invented, decorative, or absent.
Design Response + Rationale	A concrete design response is emerging, and the team can trace important design choices back to what it learned from people and prior work.	A design direction exists, but the connection between needs and design choices is still weak.	Mostly implementation or ideation without a clear human-centered rationale.
Evaluation + Method Fit	Clear what will be evaluated, against what, and why. The methods fit the question, include meaningful qualitative evidence, and use a defensible participant/sample strategy.	An evaluation plan exists, but the comparison, methods, participants, or qualitative component still need work.	Evaluation is absent, disconnected from the design, or only benchmark/metric-based.
Rigor + Path Forward	The team shows meaningful progress, understands major uncertainties and limitations, has clear ownership, and has a realistic plan to complete the project.	Some progress is visible, but rigor, ownership, or next steps remain unclear.	Mostly activity reporting with little evidence of a rigorous process or credible path forward.

Final Project Packet

The final team paper is the main intellectual product. Treat it like a submission to a dummy conference: another researcher or designer should be able to understand the problem, the people, the needs, the design response, the evaluation, and what you learned.

Component	Current expectation
Final Team Project - 35%	A **6–8 page** ACM SIGCHI-format team paper, excluding references, plus the appropriate artifact/demo. The paper should tell one coherent story: problem, people and needs, grounding in prior work, design response, evaluation, what you learned, limitations, and what should happen next.
Individual Contribution Justification - 15%	**Up to 2 pages.** Explain your intellectual contribution, consequential choices you helped make, tradeoffs, what you learned, and what you would do differently.
Peer evaluation	You will evaluate your teammates' contributions.

The detailed final-project rubric and peer-evaluation process will be shared in due time. Each student should be prepared to explain and defend their own contribution.

Need methodological help? Start with the problem, identify the method you actually need, learn enough to make a thoughtful attempt, and ask for help after doing some initial work.

FAQs (based on the 40-min live Q&A in class Oct 5)

Do we need to build a complete production-ready system? No. Build enough to express and evaluate the design idea.

Can the project mostly be interviews, surveys, or a literature review? Those can be important parts of the understanding phase, but they cannot be the entire project. What you learn should lead to a design response that you evaluate.

Do we need ideal or professional participants? No. Scope the project around people you can realistically reach. Classmates, lab mates, friends, colleagues, domain contacts, and reasonable proxies can all be appropriate when they fit the question.

Can the same people help us understand needs and later evaluate the design? Yes, when that makes sense. Going back to people who helped surface the need can be a useful way to see whether the design actually responds to it.

How sophisticated do the methods need to be? Enough to support a thoughtful, defensible attempt. Interviews do not need to be hour-long or highly formal. Learn enough to use the method responsibly and document what you did.

What if the project or evaluation fails? Show what you tried, what happened, what participants told you, and what you would change. Negative results are acceptable.

Do we need IRB approval? The course does not require every class project to obtain IRB approval. There is no requirement or expectation to publish your report. Treat it as an exercise. If you are extending an existing approved study, planning publication, or integrating the project into your ongoing research program, discuss that situation separately before proceeding.

Can we use AI tools? No, to replace human participants, conduct analysis, fabricate data, or generate "synthetic users" that you present as participant evidence. Unauthorized use of AI will trigger an academic integrity violation investigation. **Yes,** to help you learn methods, brainstorm, prototype, or orient yourself.

Does everyone on the team need to do every part? No. Teams should use complementary strengths, and responsibilities can differ. Every member must still own a load-bearing intellectual contribution and be able to explain how their work shaped the project.

Can we start from our existing research? Yes. Existing expertise or community access can be an advantage, but the project must make a genuine human-centered move. Do not simply repackage work you were already doing.

Can this become a portfolio piece or publication? Absolutely, if you plan and scope it accordingly. That is welcome, but the course requirement is a rigorous learning process, not a publication-ready result.